RISE CRM - Technical & Interview Guide

Comprehensive Architectural Breakdown & QA Automation Interview Preparation

AUT: https://rise.anhtester.com/ FRAMEWORK: Selenium Java + TestNG

1. Executive Summary & Application Overview

RISE CRM is an enterprise-grade, web-based **Customer Relationship Management (CRM) and Project Management platform**. It is built to streamline client onboarding, sales pipelines, project lifecycles, team assignments, invoicing, and support ticket management for small to medium enterprises.

Application Under Test (AUT) Specifications

- **Application URL:** `https://rise.anhtester.com/`
- **Architecture:** Single-page style responsive web app built using PHP (CodeIgniter MVC), MySQL, Bootstrap, jQuery, AJAX, and REST APIs.
- **UI Components:** Asynchronous AJAX modals, jQuery DataTables, Select2 custom dynamic dropdowns, and Chart.js dashboards.
- **Key Automation Objective:** Regression automation of critical business paths (Multi-role Authentication, Client Onboarding, Dynamic Table Filtering, and Verification).

2. Core Business Modules & Test Coverage

Module	Business Functionality	Automation Test Coverage
Authentication & RBAC	Role-Based Access Control supporting multiple user tiers: Admin, Team Member, and Client.	Login with admin/client roles, session management, negative authentication handling, URL redirection to `/dashboard` .
Dashboard (`/dashboard`)	Executive overview displaying KPI metric cards (Total Clients, Active Projects, Invoices, Open Tasks).	Post-login dashboard layout validation, menu widget presence, card text and counters.
Client Management (`/clients`)	Client directory supporting Organization and Individual client profiles, billing addresses, VAT/Tax numbers, and group categorization.	End-to-end "Add Client" workflow, Select2 dropdown selection, AJAX form submission, and database-to-UI verification.
Projects & Tasks (`/projects`)	Managing project deliverables, milestones, start/end dates, task assignments, and progress tracking.	Project creation verification, navigation between project tabs, task list filtering.
DataTables & Search	Dynamic table listing clients with server-side pagination, sorting, and real-time keyword search.	Table column text assertion, pagination traversal, dynamic search filter verification using custom table helpers.

3. Key Technical Challenges Solved (Interview Highlights)

When interviewers ask about automation challenges, describe these four specific engineering problems you solved in this framework:

1. Automating Custom Select2 jQuery Dropdowns

The Challenge: The "Owner" and "Client Groups" fields in RISE CRM do not use standard HTML `<select>` elements. Standard Selenium `Select(webElement)` throws `UnexpectedTagNameException` because they are implemented using custom `<div>` and `<span>` containers.

The Solution: Implemented a reusable keyword action:

1. Click the Select2 target container (`#s2id_created_by`).
2. Wait for the dynamic search input (`#select2-drop input`) to become visible.
3. Type the target value and send `Keys.ENTER` or click the dynamically populated match (`.select2-match`).

2. Asynchronous AJAX Modal Windows (`#ajaxModalContent`)

The Challenge: Clicking "Add Client" does not navigate to a new page; it asynchronously injects a Bootstrap modal into the DOM. Interacting with form inputs before the modal animation and AJAX call finish causes `ElementNotInteractableException` or `StaleElementReferenceException` .

The Solution: Built a smart wait helper (`WebUI.waitForPageLoaded()` and `WebUI.waitForJQueryLoad()`) that monitors `jQuery.active == 0` and `document.readyState == 'complete'` , ensuring JavaScript execution is fully idle before interacting with elements.

3. Dynamic Table Search & Column Validation

The Challenge: The client list table dynamically re-renders row contents via AJAX whenever text is typed into the search bar without triggering a browser refresh.

The Solution: Developed `checkContainsValueOnTableByColumn(int columnIndex, String expectedText)` in `WebUI.java` . It clears the search field, enters the query, waits for the DOM to settle, retrieves all column elements in the tbody, and asserts matches across table rows.

4. Secure Data-Driven Testing with Apache POI

The Challenge: Separating test logic from test data, avoiding hardcoded passwords in repositories, and testing multiple datasets.

The Solution: Used Apache POI to read records from `ClientsDataExcel.xlsx` into a `Hashtable<String, String>` via TestNG's `@DataProvider` . Passwords stored in encrypted form were decoded at runtime using `DecodeUtils.decrypt()` .

4. Framework Architecture Overview

Technology Stack & Design Patterns

- **Pattern:** Page Object Model (POM) with Page Factory and fluent method chaining.
- **Thread Safety:** Implemented Selenium's `ThreadGuard.protect()` with `ThreadLocal<WebDriver>` to enable parallel test execution across browser threads without session collisions.
- **Reports & Monitoring:** ExtentReports 5.x for interactive executive dashboards, Allure TestNG for detailed step-by-step analytics, and Log4j2 for console & file logging.
- **Failure Captures:** Automated screenshot capture on test failure hooked directly via TestNG `TestListener` (`onTestFailure`).
- **CI/CD Readiness:** Configured for headless execution (`HEADLESS=true`) for seamless integration with Jenkins pipelines.

5. High-Impact Interview Q&A (Sample Scripts)

Q1: "Can you tell me about the application you automated in your framework?"

*"I automated **RISE CRM** (https://rise.anhtester.com/), an enterprise web application designed for Client Relationship Management and Project Tracking. The system allows companies to manage the full client lifecycle—from initial lead and onboarding to project milestone tracking and invoicing. My test suite specifically covered regression paths for user authentication across multiple roles, end-to-end client creation, and search table validations. The application relies heavily on dynamic AJAX dialogs, Select2 search dropdowns, and jQuery DataTables."*

Q2: "What was the most technically challenging scenario you automated on this CRM?"

*"One of the most interesting challenges was automating the **'Add Client' modal workflow**. The form is loaded inside an asynchronous AJAX modal rather than a traditional page navigation, and several dropdowns (like Owner and Client Group) are built using custom Select2 widgets instead of native HTML select tags. Standard Selenium actions frequently failed due to timing issues. To make this robust, I implemented synchronization mechanisms that checked both document readyState and active jQuery AJAX connections (`jQuery.active === 0`), and encapsulated custom Select2 handling logic in our keyword utility class. This eliminated flakiness and made the test run seamlessly in parallel headless mode."*

Q3: "How did you manage test data and security in this framework?"

"We followed a data-driven approach using Apache POI to parse test datasets from external Excel files (`ClientsDataExcel.xlsx`). We fed this into TestNG test methods using `@DataProvider` mapped to a `Hashtable<String, String>` . To maintain security, credentials were never hardcoded in plain text—they were stored encrypted in the dataset and decoded at runtime using a decryption utility before interacting with the browser."

Q4: "How do you run these tests in CI/CD?"

"The framework is fully Maven-integrated. In our configuration, we can toggle `HEADLESS=true` in `config.properties` or pass it as a Maven command-line argument (`mvn test -DHEADLESS=true`). We can run individual TestNG suites or parallel classes. Post-execution, the framework automatically produces ExtentReports HTML dashboards, Allure test results, and archives failure screenshots for CI artifacts."

6. Quick Reference Cheat-Sheet

Item	Details
Application URL	`https://rise.anhtester.com/signin`
Admin Login	`admin@example.com` / `123456` (Redirects to `/dashboard`)
Client Login	`customer@example.com` / `123456`
Test Suite XMLs	`src/test/resources/suites/CRM/SignIn-simple.xml`, `Clients-parallel.xml`
Key Page Classes	`SignInPageCRM.java`, `DashboardPageCRM.java`, `ClientPageCRM.java`
HTML Report Path	`exports/ExtentReports/ExtentReports.html`